

1. Technical Architecture Overview

This diagram represents a standard 3-tier architecture (UI, Logic, Data) to ensure a modular and scalable design.

2. Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based interaction portal	React Js / Angular Js
2	Application Logic-1	Core request processing	Python (Flask/Django)
3	Application Logic-2	Speech-to-text processing	IBM Watson STT
4	Application Logic-3	Intelligent conversational flow	IBM Watson Assistant
5	Database	Relational data management	MySQL
6	Cloud Database	Scalable NoSQL cloud storage	IBM Cloudant
7	File Storage	Secure asset storage	IBM Block Storage

8	External API-1	Weather data integration	IBM Weather API
9	External API-2	Verification services	[Aadhaar Redacted] API
10	Machine Learning Model	Recognition and classification	Custom TensorFlow Model
11	Infrastructure	Cloud-native deployment	Kubernetes / Cloud Foundry

3. Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frontend and backend development	React, Python Flask
2	Security Implementations	Data protection & access control	SHA-256, IAM Controls, OWASP
3	Scalable Architecture	Micro-services for modular growth	Docker/Kubernetes
4	Availability	Load distribution for uptime	Load Balancers, Multi-AZ deployment

5	Performance	High-speed response optimization	Redis Cache, Cloud CDN
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Project Design Phase II: Technical Architecture Overview

